

Online Active Learning For LSTM-Based Network Traffic Prediction

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Abstract— This paper introduces an online active deep learning framework tailored for intelligent and sustainable aerial-terrestrial IoT networks. The framework utilizes Long Short-Term Memory (LSTM) models, enhanced with active learning and incremental updates, to address the challenges of dynamic network traffic prediction. Model updates are selectively applied, prioritizing samples with higher prediction errors, enabling the system to adapt dynamically to evolving traffic patterns and improving predictive accuracy. The proposed approach accommodates non-stationary data streams and prioritizes critical information, establishing a robust foundation for effective network management. The framework's performance was validated using the Milan traffic dataset, demonstrating its effectiveness compared to non-weighted online learning methods and static models.

Keywords— *LSTM networks, online learning, active learning, network traffic prediction.*

I. INTRODUCTION

The rapid proliferation of connected devices has driven the evolution of IoT networks, emphasizing the need for intelligent and sustainable solutions. Aerial-terrestrial IoT networks, which integrate airborne and ground-based components, have emerged as a transformative solution, particularly in high-demand scenarios such as disaster management, where the need for rapid, reliable, and adaptive connectivity is paramount. These networks leverage softwareization to provide operators with enhanced control over their infrastructures, enabling advanced strategies such as traffic engineering, resource allocation, and service orchestration. As highlighted by Bui et al. [1], anticipatory decision-making in network management offers the potential to significantly improve resource utilization and quality of service compared to traditional reactive methods. However, the effectiveness of these anticipatory strategies is inherently dependent on the accuracy of the underlying predictive models, positioning traffic time-series forecasting as a critical component in the design of modern IoT networks.

In the context of aerial-terrestrial IoT networks, accurate traffic forecasting is essential for ensuring reliable operation, enabling tasks such as predictive maintenance, reliability monitoring, dynamic resource allocation, and intrusion detection [2]. These capabilities are fundamental to improving availability, security, and the overall quality of service. However, the increasing prevalence of tracking and monitoring technologies has resulted in a surge of streaming data, requiring predictive models that can adapt to dynamic

environments. Conventional static models often fail to account for the dynamic and non-stationary nature of these data streams, underscoring the need for online learning methods capable of adapting to evolving traffic in real-time.

To address these challenges, this work proposes an online active deep learning framework designed to meet the specific requirements of intelligent and sustainable aerial-terrestrial IoT networks. The framework leverages Long Short-Term Memory (LSTM) models, enhanced through the integration of active learning and incremental updates. The proposed approach focuses on making model updates selective based on the informativeness of incoming data—specifically, by giving precedence to samples where prediction errors are higher. This targeted updating strategy allows the system to adapt dynamically to evolving traffic patterns, improving predictive accuracy. By accommodating non-stationary data streams and prioritizing critical information, the proposed framework establishes a robust and adaptive foundation for effective and sustainable network management.

The remaining of the paper is as follows: Section II presents the related work. The proposed methodology is presented in Section III. Section IV is devoted to experimental results and discussion. Finally, Section V closes the paper with the main conclusions extracted from this research.

II. RELATED WORK

A. Network traffic prediction

There is a wide array of network traffic prediction methods, which can be broadly categorized into classical statistical models and Artificial Neural Network (ANN) approaches [3]. Classical methods include Autoregressive Moving Average (ARMA) [4], Autoregressive Integrated Moving Average (ARIMA) [5], and Seasonal ARIMA (SARIMA) [6] models, each designed to handle different aspects of time-series data such as non-stationarity and seasonality. These methods use past traffic observations and errors to predict future trends and are based on linear models. On the other hand, Artificial Neural Networks (ANNs), such as Recurrent Neural Networks (RNNs) [7], Long Short-Term Memory (LSTM) [8] networks, and Gated Recurrent Units (GRUs) [9], are specifically designed to capture complex nonlinear patterns in sequential data. Their data-driven nature has made them the preferred choice for many applications, as they often achieve higher accuracy with reduced reliance on manual model design. However, these models come with the

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trade-off of being inherently more difficult to interpret. While eXplainable Artificial Intelligence (XAI) [10] techniques can be employed to address this limitation, achieving a comprehensive understanding of their decision-making processes remains challenging. In addition, Convolutional Neural Networks (CNNs) can be used for spatio-temporal analysis [11], and Graph Neural Networks (GNNs) [12] can be employed when network topology is relevant. Hybrid approaches, which combine different statistical and ANN methods, have also been proposed. Other machine learning techniques have also been proposed, including, Support Vector Machines (SVM) [13], and Random Forests (RF) [14], each offering different ways to approach network traffic prediction. For a more in-depth exploration of these methods, the reader is referred to a comprehensive review in [3].

B. Online learning applied to time series forecasting

Online learning algorithms process data sequentially, updating the model as new data arrives. This is in contrast to offline learning, where the model is trained on a fixed dataset [15]. Online learning is well-suited to handle data streams, which are common in real-world applications, such as network traffic prediction. It can be considered as an extension of incremental learning which updates the model example by example. Concept drift [15], a major challenge in online learning, refers to changes in the underlying data distribution over time, which can lead to a decrease in the performance of machine learning models. It occurs when the relationship between input features and the target variable evolves, making previously learned patterns obsolete or less effective. Another important challenge is catastrophic forgetting [16]: As the model learns new information, it tends to "forget" previously learned knowledge, leading to a decline in performance on older data. Both concept drift and catastrophic forgetting need to be addressed to achieve a tradeoff between stability (maintaining performance on previously seen data) of the model and its plasticity (adapting to new, non-stationary data).

Techniques like continual learning methods, such as Elastic Weight Consolidation (EWC) and Synaptic Intelligence (SI) can mitigate catastrophic forgetting [17]. EWC is based on the idea of selectively constraining weight updates for parameters that are critical to previously learned tasks. It does this by incorporating a regularization term into the loss function that penalizes significant changes to important weights. On the other hand, SI uses a different approach to measure the importance of parameters by considering the trajectory of parameter updates during training, rather than relying on the Fisher Information Matrix. Memory-based approaches, such as Experience Replay [18] and Incremental Few-Shot Learning [19], enhance model adaptability by preserving knowledge from past traffic behaviors by either keeping a memory buffer with past samples or incorporating only a few samples to the process.

Most of the recent research on online learning for time series forecasting has been focused on identifying the novelty of the incoming streaming data, so that concept drift and catastrophic forgetting can be balanced by properly considering this novelty in the learning process. Al-Falouji et al. [20] developed an anomaly detection algorithm based on a k-nearest neighbor graph, which was evaluated on both

synthetic and real-world datasets. Similarly, Carreño et al. [21] proposed an advanced novelty detection algorithm for streaming data, leveraging a mixture of Gaussian distributions to effectively capture and identify anomalies in dynamic environments. In [22], the authors use the prediction error to determine whether incoming data, is normal, novel or an anomaly to incorporate it to the online learning process accordingly. In [23], a novel adaptive gradient learning method for LSTM networks is proposed, designed for robust online time series prediction. The method addresses the challenges posed by anomalies (identified using conventional Gaussian statistics) and change points (found by assessing local differences) in streaming data by weighting gradients based on local data features. This weighting mechanism differentiates between outliers, which should be downweighted to prevent model drift, and change points, which necessitate quick model adaptation.

III. METHODOLOGY

A. Online learning for time series prediction

We denote a vector time series by $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_T\}$, where the subscript t represents a time stamp and each element in the series $\mathbf{x}_t$ is a $N \times 1$ vector. The predicted new data point $\hat{\mathbf{x}}_t$ is obtained from past values as:

$$\hat{\mathbf{x}}_t = g(\mathbf{x}_{t-1}, \mathbf{x}_{t-2}, \dots, \mathbf{x}_1, \theta_{t-1}^*) \quad (1)$$

where $g(\cdot)$ is a continuous function modelling the dynamic evolution of the time series. In this paper $g(\cdot)$ is obtained using a LSTM network, a type of RNN designed to address the issue of vanishing gradients, which can make it difficult for standard RNNs to learn long-term dependencies in data. θ_{t-1}^* denote the parameters of the LSTM learned with the $t-1$ past observations. Unlike neurons in traditional RNNs, which overwrite their content at each time step [24], an LSTM neuron can decide whether to retain existing memory using its gate mechanisms. Intuitively, if the neuron identifies a significant feature in an input sequence early on, it can preserve this information (representing the feature's presence) over extended periods. This ability enables LSTMs to effectively capture hidden long-term dependencies [25].

As a new sample $\mathbf{x}_t$ arrives, the model is updated by minimizing a loss function:

$$\theta_t^* = \underset{\theta_t}{\operatorname{argmin}} E(\mathbf{x}_t, g(\mathbf{x}_{t-1}, \mathbf{x}_{t-2}, \dots, \mathbf{x}_1, \theta_{t-1}^*)) \quad (2)$$

with $E(\cdot, \cdot)$ an error function such as the Mean Squared Error (MSE) or the Mean Absolute Error (MAE). In this paper, we have chosen the MSE as loss function:

$$\theta_t^* = \underset{\theta_t}{\operatorname{argmin}} (\mathbf{x}_t - g(\mathbf{x}_{t-1}, \mathbf{x}_{t-2}, \dots, \mathbf{x}_1, \theta_{t-1}^*))^2 \quad (3)$$

The new parameters θ_t^* are updated recursively in the online learning process to predict the next data sample $\hat{\mathbf{x}}_{t+1}$. To further enhance the efficiency of online learning, active learning [26] can be integrated into this process. By incorporating active learning, the most informative examples from the data stream can be prioritized in the incremental learning process. The online adaptive learning procedure, with an active learning extension, involves the following steps:

1. Predict $\hat{\mathbf{x}}_t$: Make a prediction using the current model for the new input data using (1).
2. Diagnose: Evaluate the loss after receiving the true value $E(\mathbf{x}_t, g(\mathbf{x}_{t-1}, \mathbf{x}_{t-2}, \dots, \mathbf{x}_1, \theta_{t-1}^*))$.
3. Determine the degree of uncertainty/informativeness of the new sample from the previous diagnosis. We will use the actual error to evaluate this.
4. Update: Use the obtained information to update the model with the new sample (2). Accurate predictions are likely to be less informative than non-accurate ones. Thus, uncertain predictions will contribute more significantly to the update.

B. Online active learning of LSTM networks using weighted gradients

Fig. 1 shows a scheme of the operations inside a LSTM unit. The involved parameters are described as follows:

- $\mathbf{h}_t \in \mathbb{R}^{p \times 1}$ is the hidden state at time t . It is obtained as the element-wise product:

$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh \mathbf{c}_t \quad (4)$$

- $\mathbf{o}_t \in \mathbb{R}^{p \times 1}$ is the output gate that modulates the amount of memory content exposure:

$$\mathbf{o}_t = \sigma(\mathbf{W}_o \mathbf{x}_t + \mathbf{U}_o \mathbf{h}_{t-1} + \mathbf{V}_o \mathbf{c}_t + \mathbf{b}_o) \quad (5)$$

where σ is the sigmoid function, $\mathbf{W}_o \in \mathbb{R}^{p \times N}$, $\mathbf{U}_o \in \mathbb{R}^{p \times p}$, $\mathbf{V}_o \in \mathbb{R}^{p \times p}$ are weight matrices to be obtained during the training process, the latter, a diagonal matrix, and $\mathbf{b}_o \in \mathbb{R}^{p \times 1}$ is a trainable bias vector.

- $\mathbf{c}_t \in \mathbb{R}^{p \times 1}$ is the memory cell content, which is updated in such a way that part of the previous memory is forgotten and replaced by new memory content obtained from the input and the current hidden state:

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t \quad (6)$$

$$\tilde{\mathbf{c}}_t = \tanh(\mathbf{W}_c \mathbf{x}_t + \mathbf{U}_c \mathbf{h}_{t-1} + \mathbf{b}_c) \quad (7)$$

where, again, $\mathbf{W}_c \in \mathbb{R}^{p \times N}$, $\mathbf{U}_c \in \mathbb{R}^{p \times p}$ are weight matrices to be obtained, and $\mathbf{b}_c \in \mathbb{R}^{p \times 1}$ is a

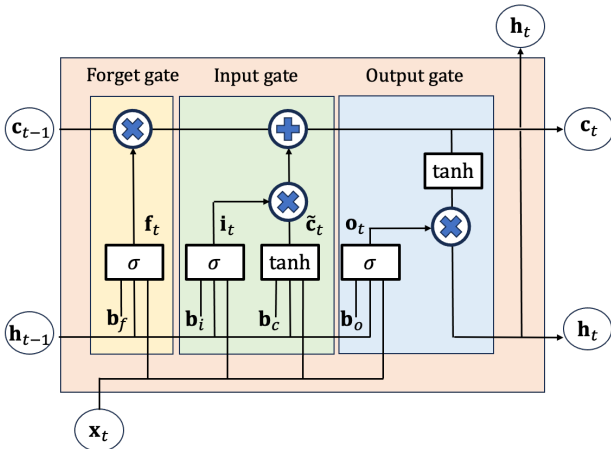


Fig. 1. Scheme of the operations inside a LSTM unit.

trainable bias vector.

- The forget $\mathbf{f}_t \in \mathbb{R}^{p \times 1}$ and input $\mathbf{i}_t \in \mathbb{R}^{p \times 1}$ gates are obtained respectively as:

$$\mathbf{f}_t = \sigma(\mathbf{W}_f \mathbf{x}_t + \mathbf{U}_f \mathbf{h}_{t-1} + \mathbf{b}_f) \quad (7)$$

$$\mathbf{i}_t = \sigma(\mathbf{W}_i \mathbf{x}_t + \mathbf{U}_i \mathbf{h}_{t-1} + \mathbf{b}_i) \quad (8)$$

with the same considerations for weight matrices and bias vectors as above.

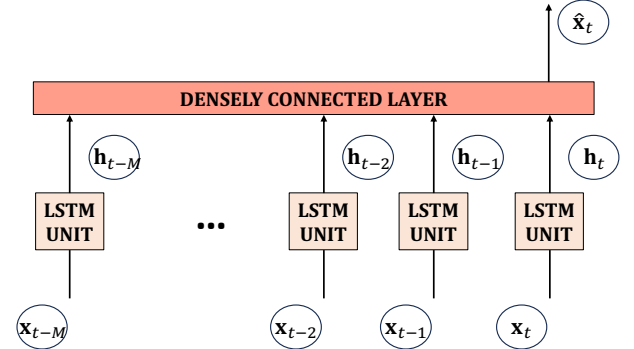


Fig. 2. Implemented LSTM network.

Fig. 2 shows the complete implemented network where M LSTM units are connected to form the LSTM layer. Its outputs are mapped to the target prediction through a dense layer. The value of $\hat{\mathbf{x}}_t$ is finally obtained as:

$$\hat{\mathbf{x}}_t = g(\mathbf{W}_d \mathbf{h}_t^s + \mathbf{b}_d) \quad (8)$$

where $\mathbf{W}_d \in \mathbb{R}^{N \times (Mp)}$ and $\mathbf{b}_d \in \mathbb{R}^{N \times 1}$ are the dense connected layer weight matrix and bias vector respectively, and $\mathbf{h}_t^s \in \mathbb{R}^{(Mp) \times 1}$ is a stacked vector concatenating each neuron's hidden state vectors. We denote by $g(\cdot)$ the activation function (sigmoid, tanh, linear, etc.) of the dense layer in this case.

Denoting by $\bar{\mathbf{W}}_t$ the set of all trainable parameters at time t :

$\bar{\mathbf{W}}_t = \langle \mathbf{W}_o, \mathbf{U}_o, \mathbf{V}_o, \mathbf{b}_o, \mathbf{W}_c, \mathbf{U}_c, \mathbf{b}_c, \mathbf{W}_f, \mathbf{U}_f, \mathbf{b}_f, \mathbf{W}_i, \mathbf{U}_i, \mathbf{b}_i, \mathbf{W}_d, \mathbf{b}_d \rangle_t$, online learning continuously updates $\bar{\mathbf{W}}_t$ to minimize a loss function. In order to do that, an iterative gradient descent procedure is followed [25]:

$$\bar{\mathbf{W}}_t = \bar{\mathbf{W}}_{t-1} - \beta \nabla E_t(\bar{\mathbf{W}}_{t-1}) \quad (9)$$

where β is the learning rate, $E_t(\bar{\mathbf{W}}_{t-1}) = (\mathbf{x}_t - \hat{\mathbf{x}}_t)^2$ is the error function (squared error in our case), and ∇ denotes the gradient operator. Taking the set of trainable parameters into (1)

$$\hat{\mathbf{x}}_t = g(\mathbf{x}_{t-1}, \mathbf{x}_{t-2}, \dots, \mathbf{x}_1, \bar{\mathbf{W}}_{t-1}), \quad (10)$$

it can be demonstrated that

$$\nabla E_t(\bar{\mathbf{W}}_{t-1}) = -2(\mathbf{x}_t - \hat{\mathbf{x}}_t) \nabla g(\mathbf{x}_{t-1}, \mathbf{x}_{t-2}, \dots, \mathbf{x}_1, \bar{\mathbf{W}}_{t-1}) \quad (11)$$

where the derivation of $\nabla g(\mathbf{x}_{t-1}, \mathbf{x}_{t-2}, \dots, \mathbf{x}_1, \bar{\mathbf{W}}_{t-1})$ can be found in [25].

To prioritize highly informative samples during the updating process, we propose weighting the gradient by the Euclidean squared norm of the normalized prediction error. Accordingly, (9) is modified as follows:

$$\bar{\mathbf{W}}_t = \bar{\mathbf{W}}_{t-1} - \beta \left\| \frac{\mathbf{x}_t - \hat{\mathbf{x}}_t}{\mathbf{x}_t} \right\|^2 \nabla E_t(\bar{\mathbf{W}}_{t-1}). \quad (12)$$

This approach ensures that uncertain predictions with larger errors contribute more significantly to model updates, while accurate predictions with smaller errors have a reduced impact.

IV. RESULTS AND DISCUSSION

Our approach was validated over the Milan traffic dataset [27]. The dataset contains hourly aggregated telecommunication and weather data from Milan, covering the period from November 1st, 2013, to January 1st, 2014. Telecommunication data includes Call Detail Records (CDRs) for received and sent SMS, incoming and outgoing calls, and internet activity, while weather data includes humidity, temperature, and wind speed. The data is hourly aggregated over 300 m² areas within the city, represented by a 1x1 m grid, with a binary adjacency matrix illustrating spatial connectivity. We focused only on the internet activity data of three separate areas, identified as pixels 4259, 4456, and 5060.

We designed a simple neural network architecture with one hidden layer containing 24 LSTM units, followed by a dense layer (see Fig. 2). The activation functions used were tanh for the LSTM layer and linear for the dense layer. A pre-trained version of the model was initialized using 50% of the data samples (700), while the remaining 700 samples were reserved for testing the online learning approach. The pre-training phase involved 400 epochs with a batch size of 25 samples, using the Adam optimizer with a learning rate of 0.001 and the mean squared error (MSE) loss function. Online training was conducted each sample for 5 additional epochs, keeping the same hyperparameters. The model's performance was evaluated using the Normalized Root Mean Squared Error (NRMSE), where the RMSE was normalized by the standard deviation of the ground truth values.

For the sake of comparison, we evaluated the performance for non-gradient-weighted online learning (NGWOL), i.e., the model was updated with every sample, but no error-dependent priority was given to any of them. Figure 3 shows the predictions obtained from our method (gradient-weighted online learning, GWOL) and NGWOL over the test dataset. The actual traffic and predictions from the pre-trained model (with no online learning) are also presented. Performance metrics for both methods are compared in Table I. Prediction results over the training and test data (with no online update of the model) are also presented for reference.

TABLE I. OBTAINED FORECASTING ERRORS

Region	NRMSE			
	Training set	Test set (no online learning)	GWOL	NGWOL
4259	0.1729	0.2982	0.1778	0.1932
4456	0.2607	0.4039	0.2455	0.2723
5060	0.1820	0.2452	0.1964	0.2125

The figure and table show that the predictions generated by the online learning methods are significantly more accurate than those produced by the offline pre-trained model. In particular, our GWOL method demonstrates a

strong tendency to converge toward the NRMSE achieved on the training set, underscoring its adaptive capabilities and superior performance. Moreover, the error figures provide clear evidence that our GWOL strategy outperforms the alternative online learning approach, further validating its

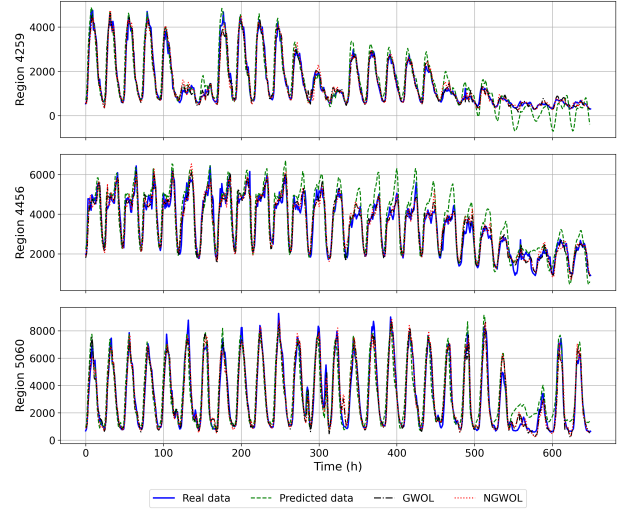


Fig. 3. Comparison of prediction results for the offline pretrained model, GWOL and NGWOL. Actual data is presented for reference.

effectiveness.

V. CONCLUSION

We presented an online active learning framework for LSTM networks designed to address the challenges of traffic prediction in aerial-terrestrial IoT networks. The method prioritizes the learning process by weighting the gradients by the normalized Euclidean squared norm of the prediction error. This approach allows for efficient model adaptation to non-stationary data streams by focusing on informative samples. The validation of the proposed method using the Milan traffic dataset demonstrates the superior performance of our GWOL strategy compared to alternative online learning approaches, and the offline pre-trained model. Our proposed method provides a robust and effective way to maintain the stability of the model while adapting to new non-stationary data, paving the way to dynamic and resilient traffic management in UAV-enabled IoT networks.

ACKNOWLEDGMENT

This work was supported by the EU Horizon 2020 Research and Innovation Programme under the Marie Skłodowska-Curie grant agreement No 101008297. This article reflects only the authors' view. The European Union Commission is not responsible for any use that may be made of the information it contains. The research was also partially supported by project TED2021-131536B-I00, funded by Spanish MCIN/AEI/10.13039/501100011033 and EU's "NextGenerationEU"/PRTR.

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